DESIGN AND RTL-TO-GDSII IMPLEMENTATION OF A SINGLE-CYCLE RISC-V PROCESSOR

INTERNSHIP REPORT

submitted by

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to

College of Engineering Trivandrum
in partial fulfillment of the requirements for the successful completion of the

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under the

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Chips-to-Startup (C2S) Programme



DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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JULY 2026

DECLARATION

I, **Rahul R** (Reg. No. **TVE23AE049**), hereby declare that the internship report entitled **"DESIGN AND RTL-TO-GDSII IMPLEMENTATION OF A SINGLE-CYCLE RISC-V PROCESSOR"** is a bonafide record of the internship work carried out by me under the **Chips-to-Startup (C2S) Programme**, sponsored by the **Ministry of Electronics and Information Technology (MeitY), Government of India**, under the guidance of **Dr. Nikhil M**, Assistant Professor, Department of Electronics and Communication Engineering, College of Engineering Trivandrum.

I further declare that this report is my original work and has not been submitted, either in whole or in part, to any university or institution for the award of any degree, diploma or other academic qualification. Wherever the work of others has been referred to, appropriate acknowledgement has been made. I affirm that this report has been prepared in accordance with the principles of academic honesty and integrity.

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Date : July 2026

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
COLLEGE OF ENGINEERING TRIVANDRUM



CERTIFICATE

This is to certify that the internship report entitled **"DESIGN AND RTL-TO-GDSII IMPLEMENTATION OF A SINGLE-CYCLE RISC-V PROCESSOR"**, submitted by **Rahul R (Reg. No. TVE23AE049)**, is a bonafide record of the internship work carried out under the **Chips-to-Startup (C2S) Programme**, sponsored by the **Ministry of Electronics and Information Technology (MeitY), Government of India**, at the **Department of Electronics and Communication Engineering, College of Engineering Trivandrum**.

This report is submitted as part of the requirements of the **C2S Internship Programme** and has not been submitted, either in whole or in part, for any other academic award.

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ABSTRACT

The RTL-to-GDSII design flow forms the backbone of modern Application-Specific Integrated Circuit (ASIC) implementation by transforming a Register Transfer Level (RTL) description into a manufacturable integrated circuit layout. Understanding this complete implementation flow is essential for developing practical knowledge of digital VLSI design and semiconductor technology.

This internship was carried out under the **Chips-to-Startup (C2S) Programme**, sponsored by the **Ministry of Electronics and Information Technology (MeitY), Government of India**, at the Department of Electronics and Communication Engineering, College of Engineering Trivandrum. The objective of the internship was to understand and implement the complete RTL-to-GDSII ASIC design flow using industry-standard Cadence Electronic Design Automation (EDA) tools.

A Single-Cycle RISC-V Processor was designed using Verilog Hardware Description Language (HDL) and verified through RTL simulation. The verified RTL design was synthesized into a technology-mapped gate-level netlist using Cadence Genus. The synthesized implementation was validated through Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS). The design was subsequently implemented using Cadence Innovus through floorplanning, power planning, placement, clock tree synthesis, routing, timing analysis, and physical verification before generating the final GDSII layout.

The successful completion of every stage of the implementation demonstrated the correctness of the processor design and provided valuable practical exposure to the industrial ASIC implementation flow. The internship significantly strengthened the understanding of digital integrated circuit design, physical implementation, timing closure, verification methodologies, and layout generation using professional EDA tools.

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List of Abbreviations

Abbreviation	**Meaning**
ASIC	Application-Specific Integrated Circuit
RTL	Register Transfer Level
GDSII	Graphic Data System II
HDL	Hardware Description Language
ISA	Instruction Set Architecture
CPU	Central Processing Unit
ALU	Arithmetic Logic Unit
EDA	Electronic Design Automation
LEC	Logic Equivalence Checking
GLS	Gate-Level Simulation
CTS	Clock Tree Synthesis
STA	Static Timing Analysis
DRC	Design Rule Check
LVS	Layout Versus Schematic
P&R	Place and Route
PDK	Process Design Kit
SDC	Synopsys Design Constraints
I/O	Input/Output
WNS	Worst Negative Slack
TNS	Total Negative Slack

Chapter 1

INTRODUCTION

1.1 Background

Advances in semiconductor technology have enabled the development of highly sophisticated digital integrated circuits capable of delivering greater computational performance while maintaining low power consumption and efficient silicon utilization. Modern electronic applications, including communication systems, consumer electronics, automotive systems, industrial automation, healthcare devices, and embedded platforms, require specialized hardware that can satisfy demanding performance, reliability, and efficiency requirements. As a result, the use of Application-Specific Integrated Circuits (ASICs) has grown significantly because they provide optimized solutions for dedicated applications compared to programmable hardware.

The implementation of an ASIC follows a systematic process known as the RTL-to-GDSII design flow. This process begins with the creation of a Register Transfer Level (RTL) description using a Hardware Description Language (HDL). The design is then validated through functional simulation, followed by logic synthesis, physical design, timing verification, and physical validation before producing the final GDSII layout used for chip fabrication. Every stage in this flow is essential to ensure that the integrated circuit satisfies functional, timing, power, and manufacturing requirements.

With the growing complexity of integrated circuit designs, Electronic Design Automation (EDA) tools have become indispensable in modern semiconductor development. These tools automate various stages of the design process, helping designers improve productivity, reduce implementation time, and achieve high-quality results. From synthesis and physical implementation to verification and timing analysis, EDA tools play a vital role throughout the complete RTL-to-GDSII flow.

This internship was conducted under the **Chips-to-Startup (C2S) Programme**, sponsored by the **Ministry of Electronics and Information Technology (MeitY),**

Government of India, at the Department of Electronics and Communication Engineering, College of Engineering Trivandrum. The primary objective was to understand and implement the complete RTL-to-GDSII ASIC design flow using industry-standard Cadence EDA tools. A **Single-Cycle RISC-V Processor** was designed in Verilog, verified through simulation, synthesized, physically implemented, and analyzed before generating the final GDSII layout.

The internship provided valuable hands-on experience in the complete ASIC implementation process and enhanced the understanding of processor architecture, digital VLSI design, physical design techniques, timing analysis, and verification methodologies used in the semiconductor industry..

1.2 Objectives

The primary objectives of this internship are as follows.

- To gain a comprehensive understanding of the complete RTL-to-GDSII ASIC design and implementation flow.
- To study the architecture, design principles, and operation of a Single-Cycle RISC-V Processor.
- To develop the processor at the Register Transfer Level (RTL) using Verilog Hardware Description Language (HDL).
- To validate the RTL design through functional simulation and verification.
- To perform logic synthesis and generate a technology-mapped gate-level netlist using Cadence Genus.
- To verify the synthesized design using Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS).
- To carry out the physical design process, including floorplanning, power planning, placement, Clock Tree Synthesis (CTS), and routing using Cadence Innovus.
- To analyze the timing performance of the implemented design through Static Timing Analysis (STA) using Cadence Tempus.
- To complete physical verification and produce the final GDSII layout suitable for fabrication.
- To acquire practical exposure to industry-standard Electronic Design Automation (EDA) tools and the overall ASIC implementation workflow.

1.3 Organization of the Report

This internship report is divided into five chapters, each covering a specific aspect of the work carried out during the internship.

Chapter 1 provides an introduction to the internship by presenting the background of ASIC design, highlighting the significance of the RTL-to-GDSII design flow, stating the objectives of the internship, and outlining the structure of the report.

Chapter 2 discusses the theoretical concepts required for the work, including the ASIC implementation flow, the RISC-V Instruction Set Architecture (ISA), the architecture of the Single-Cycle RISC-V Processor, and the Cadence Electronic Design Automation (EDA) tools used during the implementation.

Chapter 3 explains the methodology adopted to complete the internship. It describes each stage of the RTL-to-GDSII flow, beginning with RTL design and functional verification, followed by logic synthesis, physical design, timing analysis, and physical verification.

Chapter 4 presents the implementation results obtained at every stage of the design flow. It includes the outcomes of RTL simulation, synthesis, verification, physical implementation, timing analysis, and the final GDSII generation, along with a discussion of the results.

Chapter 5 summarizes the overall work completed during the internship, highlights the knowledge gained from the implementation process, and discusses possible future improvements and extensions to the implemented Single-Cycle RISC-V Processor.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

Application-Specific Integrated Circuits (ASICs) play a significant role in modern digital systems by providing high performance, low power consumption, enhanced reliability, and efficient utilization of silicon area. Unlike programmable hardware devices, ASICs are custom-designed to perform specific tasks, enabling better optimization in terms of speed, area, and energy efficiency. Owing to these advantages, ASICs are widely employed in communication networks, consumer electronics, automotive systems, industrial automation, embedded applications, artificial intelligence accelerators, and Internet of Things (IoT) devices.

The design and implementation of an ASIC are carried out through a well-defined process known as the RTL-to-GDSII design flow. This flow begins with a Register Transfer Level (RTL) description written in a Hardware Description Language (HDL) and progresses through several stages, including functional verification, logic synthesis, physical implementation, timing analysis, and physical verification, before producing the final GDSII layout for fabrication. Each stage is essential to ensure that the design satisfies the required functional, timing, power, and manufacturing specifications.

This chapter provides the theoretical foundation for the work carried out during the internship. It introduces the RTL-to-GDSII ASIC implementation flow, discusses the RISC-V Instruction Set Architecture (ISA) and the Single-Cycle RISC-V Processor, and presents an overview of the Cadence Electronic Design Automation (EDA) tools utilized throughout the implementation process.

2.2 RTL-to-GDSII Design Flow

The RTL-to-GDSII design flow is the widely accepted methodology used in the semiconductor industry for implementing digital integrated circuits. The process starts with describing the intended hardware functionality at the Register Transfer Level (RTL) using a Hardware Description Language (HDL) such as Verilog. At this stage, the design focuses on the logical behaviour of the circuit without considering its physical implementation.

Once the RTL design is completed, it undergoes functional verification through simulation to ensure that the circuit operates according to the design specifications. Detecting and correcting errors during the RTL stage is highly advantageous, as modifications become more difficult and time-consuming during the later stages of implementation.

After successful verification, the RTL design is synthesized into a technology-mapped gate-level netlist using the standard-cell library of the target fabrication technology. During this stage, timing constraints defined in the Synopsys Design Constraints (SDC) file guide the synthesis tool in optimizing the design to achieve the desired performance, area, and timing requirements.

The synthesized netlist is then validated using Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS) to ensure that the synthesis process has preserved the functionality of the original RTL description. Once the functional equivalence of the design is confirmed, it advances to the physical design stage.

The physical implementation process includes floorplanning, power planning, standard-cell placement, Clock Tree Synthesis (CTS), routing, design optimization, Static Timing Analysis (STA), and physical verification. After successfully completing these stages, the final chip layout is exported as a GDSII file, which serves as the layout database used for semiconductor fabrication.

2.3 RISC-V Instruction Set Architecture

RISC-V is an open-standard Reduced Instruction Set Computer (RISC) Instruction Set Architecture (ISA) developed at the University of California, Berkeley. Unlike proprietary processor architectures, RISC-V is freely available for academic, research, and commercial applications without licensing restrictions. Its open architecture has significantly contributed to its widespread adoption in modern processor design.

The RISC-V architecture follows a modular design philosophy consisting of a small base instruction set together with optional standard extensions supporting multiplication, floating-point arithmetic, compressed instructions, atomic operations, vector processing,

and other advanced capabilities. This modularity allows designers to implement processors according to the requirements of different applications while maintaining software compatibility.

Because of its simplicity, scalability, and flexibility, RISC-V has become one of the most popular processor architectures for embedded systems, Internet of Things (IoT) devices, artificial intelligence accelerators, edge computing platforms, and educational processor development.

2.4 Single-Cycle RISC-V Processor

A Single-Cycle RISC-V Processor is designed such that every instruction is executed within a single clock cycle. During each clock period, all stages of instruction execution—including instruction fetch, instruction decode, execution, memory access, and register write-back—are completed before the arrival of the next clock pulse. Although this approach requires the clock period to be long enough to accommodate the instruction with the maximum execution time, it offers a straightforward datapath and control unit, making it well suited for understanding processor architecture and digital system design.

The processor implemented in this internship comprises several essential hardware modules, including the Program Counter (PC), Instruction Memory, Register File, Arithmetic Logic Unit (ALU), Main Control Unit, ALU Decoder, Immediate Generator, Data Memory, multiplexers, and additional supporting control logic. These modules work together to execute arithmetic, logical, memory access, and branch instructions as defined by the RISC-V Instruction Set Architecture (ISA).

The modular design of the processor improves code organization, simplifies debugging and functional verification, and allows future enhancements to be incorporated with ease. Owing to its simple architecture and ease of implementation, the Single-Cycle RISC-V Processor is widely used for educational purposes, introductory processor design, and learning the complete RTL-to-GDSII ASIC implementation flow.

2.5 Electronic Design Automation Tools

The design and implementation of modern integrated circuits involve highly complex processes that cannot be performed efficiently without the support of Electronic Design Automation (EDA) tools. These software tools automate various stages of the ASIC design flow, including synthesis, verification, physical implementation, timing analysis, and layout generation. By automating these tasks, EDA tools improve design accuracy, reduce development time, and help achieve better performance, power, and area optimization.

The following Cadence EDA tools were utilized during the internship.

- **Cadence Genus** was used to convert the Verilog RTL design into a technology-mapped gate-level netlist through logic synthesis.
- **Cadence Conformal LEC** was employed to perform Logic Equivalence Checking (LEC) between the RTL design and the synthesized netlist, as well as between the synthesized and post-route implementations.
- **Cadence Innovus** was used to carry out the physical design stages, including floorplanning, power planning, placement, Clock Tree Synthesis (CTS), routing, and physical optimization.
- **Cadence Tempus** was utilized for Static Timing Analysis (STA) to verify setup and hold timing requirements and perform timing signoff.
- **Cadence Virtuoso** was used to visualize and inspect the final physical layout generated in the GDSII format.

The combined use of these industry-standard EDA tools enabled the successful completion of the RTL-to-GDSII implementation of the Single-Cycle RISC-V Processor while ensuring functional correctness, timing closure, and a reliable physical design suitable for fabrication.

2.6 Summary

This chapter provided the theoretical background required for understanding the work carried out during the internship. It covered the RTL-to-GDSII ASIC design flow, the RISC-V Instruction Set Architecture (ISA), the architecture of the Single-Cycle RISC-V Processor, and an overview of the Cadence Electronic Design Automation (EDA) tools employed during the implementation. The concepts discussed in this chapter establish the necessary foundation for understanding the implementation methodology described in the next chapter.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter explains the methodology followed during the internship to implement the Single-Cycle RISC-V Processor using the complete RTL-to-GDSII ASIC design flow. The implementation was carried out by following the standard ASIC design methodology adopted in the semiconductor industry, where each stage was thoroughly completed and verified before moving to the next phase. This step-by-step approach ensured the functional correctness of the design throughout the implementation process and minimized the chances of errors during the later stages of physical design.

The entire implementation was performed using industry-standard Cadence Electronic Design Automation (EDA) tools as part of the Chips-to-Startup (C2S) Programme. Starting with the Register Transfer Level (RTL) design developed in Verilog Hardware Description Language (HDL), the processor underwent functional verification, logic synthesis, post-synthesis validation, physical design implementation, timing analysis, physical verification, and finally GDSII generation. The output generated at each stage served as the input for the next stage, resulting in a systematic and complete RTL-to-GDSII implementation flow.

3.2 Overall Design Flow

The implementation process followed during the internship was carried out through a sequence of interconnected stages, beginning with RTL development and ending with the generation of the final GDSII layout. The major stages involved in the design flow are listed below.

1. RTL Modelling of the Processor

2. RTL Simulation and Functional Validation
3. Technology Mapping through Logic Synthesis
4. Post-Synthesis Functional Verification
5. Floorplanning, Placement, CTS and Routing
6. Timing Verification and Signoff
7. Physical Design Verification
8. GDSII File Generation

The successful completion and verification of each stage ensured that the processor maintained its functional correctness and met the implementation requirements before proceeding to the next phase of the ASIC design flow.

3.3 RTL Modelling of the Processor

The implementation process commenced with the RTL modelling of the Single-Cycle RISC-V Processor using Verilog Hardware Description Language (HDL). The processor architecture was developed by designing individual functional modules, each responsible for a specific operation within the processor. These modules included the Program Counter (PC), Instruction Memory, Register File, Arithmetic Logic Unit (ALU), Main Control Unit, ALU Decoder, Immediate Generation Unit, Data Memory, multiplexers, and other supporting logic required for proper processor functionality.

After the successful development of the individual modules, they were integrated to form the complete processor architecture. A modular design approach was adopted to improve code organization, simplify debugging, facilitate independent testing of each hardware block, and make future modifications easier. The finalized RTL design served as the foundation for the subsequent stages of the RTL-to-GDSII implementation flow.

3.4 RTL Simulation and Functional Validation

Following the completion of the RTL modelling phase, the processor design was validated through functional simulation using a Verilog testbench. A variety of instruction sequences were executed to evaluate the correct operation of arithmetic, logical, load/store, and branch instructions supported by the implemented Single-Cycle RISC-V Processor.

The simulation results were carefully analyzed and compared with the expected outputs to confirm the functional behaviour of the processor. The successful completion of

RTL simulation verified that the design operated as intended and satisfied the required functional specifications, making it ready for the logic synthesis stage.

3.5 Technology Mapping through Logic Synthesis

After the successful completion of RTL simulation and functional validation, the processor design proceeded to the logic synthesis stage using Cadence Genus. During this phase, the RTL description written in Verilog was converted into a technology-mapped gate-level netlist by utilizing the standard-cell library of the selected semiconductor technology.

The synthesis process was guided by timing constraints specified in the Synopsys Design Constraints (SDC) file. Based on these constraints, the synthesis tool optimized the design to achieve the desired timing, area, and performance objectives. Upon completion, various reports containing information on timing performance, area utilization, design hierarchy, and standard-cell usage were generated for further analysis and verification.

3.6 Verification of the Synthesized Netlist

Before proceeding to the physical design stage, the synthesized gate-level netlist was thoroughly verified to ensure that the logic synthesis process had not altered the intended functionality of the RTL design.

The synthesized netlist was first validated using Cadence Conformal through Logic Equivalence Checking (LEC), where it was compared with the original RTL description. Subsequently, Gate-Level Simulation (GLS) was performed using the synthesized netlist along with the same testbench employed during RTL simulation. The successful completion of both LEC and GLS confirmed that the synthesized design was functionally equivalent to the RTL implementation and was ready for physical design implementation.

3.7 Physical Design using Cadence Innovus

After the successful verification of the synthesized netlist, the design was imported into Cadence Innovus to begin the physical design stage. This phase converted the synthesized logical design into a physical layout that could be manufactured using semiconductor fabrication technology.

The physical implementation started with floorplanning, where the die size, core region, placement rows, and I/O pad locations were defined to establish the overall chip layout. This was followed by power planning, during which power rings and power stripes

were created to provide a stable and efficient power distribution network across the entire design.

Once the power network was established, the synthesized standard cells were placed within the core area while optimizing wire length, congestion, and overall layout utilization. Clock Tree Synthesis (CTS) was then performed to distribute the clock signal efficiently to all sequential elements, minimizing clock skew and insertion delay. Finally, the routing stage established all signal, clock, and power interconnections in accordance with the design rules and routing constraints of the target technology.

3.8 Static Timing Analysis and Timing Signoff

After completing the routing stage, Static Timing Analysis (STA) was carried out using Cadence Tempus to evaluate the timing characteristics of the implemented processor. Detailed timing reports were generated to verify that the design satisfied the required setup and hold timing constraints under the specified operating conditions.

The critical timing paths were examined carefully to ensure that the processor could operate at the target clock frequency without any timing violations. Wherever required, timing optimization was performed to improve the design before moving to the physical verification stage, thereby achieving successful timing signoff.

3.9 Physical Design Verification

After the completion of the physical design stage, the generated layout was subjected to physical verification to ensure that it satisfied the manufacturing requirements of the selected semiconductor technology.

Design Rule Checking (DRC) was performed to examine the layout for violations related to spacing, width, enclosure, overlap, and other fabrication design rules. In addition, Post-Route Logic Equivalence Checking (LEC) was carried out to verify that the routing and physical optimization processes had not affected the logical behaviour of the processor. The successful completion of these verification steps confirmed that the final layout was functionally correct, free from design rule violations, and suitable for semiconductor fabrication.

3.10 Summary

This chapter presented the methodology adopted to implement the Single-Cycle RISC-V Processor by following the complete RTL-to-GDSII ASIC design flow. The implementa-

tion commenced with RTL modelling of the processor, followed by RTL simulation and functional validation, technology mapping through logic synthesis, verification of the synthesized netlist, physical design using Cadence Innovus, static timing analysis and timing signoff, and physical design verification before generating the final GDSII layout. The successful completion and validation of each stage ensured that the processor maintained functional correctness and met the requirements for successful ASIC implementation and fabrication.

Chapter 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the outcomes obtained from each stage of the RTL-to-GDSII implementation of the Single-Cycle RISC-V Processor. Every stage of the design flow was successfully completed and verified before proceeding to the next phase, ensuring the functional correctness and reliability of the implemented processor. The results obtained from RTL simulation, logic synthesis, post-synthesis verification, physical design, timing analysis, and physical verification demonstrate the successful implementation of the processor using the standard ASIC design methodology.

Unlike the previous chapter, which focused on the implementation procedure, this chapter discusses the results generated during each stage of the design flow and highlights their significance. The accompanying figures provide visual evidence of the implementation process and support the observations presented throughout this chapter.

4.2 RTL Implementation and Verification Results

The Single-Cycle RISC-V Processor was successfully modelled at the Register Transfer Level (RTL) using Verilog Hardware Description Language (HDL). The complete processor architecture was developed by integrating several functional modules, including the Program Counter (PC), Instruction Memory, Register File, Arithmetic Logic Unit (ALU), Main Control Unit, ALU Decoder, Immediate Generation Unit, Data Memory, multiplexers, and supporting control logic. The modular design approach simplified development, enabled independent testing of individual modules, and facilitated the successful integration of the complete processor.

Figure 4.1 illustrates the RTL architecture of the implemented processor.

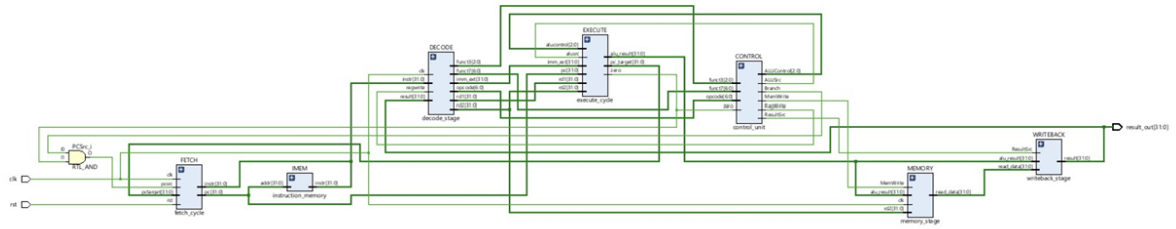


Figure 4.1: Block Diagram of the Single-Cycle RISC-V Processor

The RTL architecture clearly depicts the datapath and control path of the processor along with the interaction between its functional modules. The successful integration of these modules confirms that the processor architecture has been implemented according to the RISC-V Instruction Set Architecture (ISA). The modular organization also simplified debugging, verification, and future design enhancements.

Functional validation of the RTL design was carried out using a Verilog testbench. A variety of instruction sequences were executed to verify the correct operation of arithmetic, logical, load/store, and branch instructions supported by the implemented processor.

The simulation results confirm that the processor executed all the applied test cases correctly. The observed register updates, ALU outputs, memory operations, and control signals matched the expected behaviour throughout the simulation. The successful completion of RTL simulation verified the functional correctness of the processor and confirmed its readiness for the logic synthesis stage.

4.3 Synthesis Results

Following successful RTL simulation and functional validation, the processor design was synthesized using Cadence Genus. During this stage, the Verilog RTL description was translated into a technology-mapped gate-level netlist using the standard-cell library of the selected semiconductor technology. Timing constraints defined through the Synopsys Design Constraints (SDC) file guided the synthesis process to achieve the required timing objectives.

Figure 4.2 illustrates the synthesis process performed using Cadence Genus.

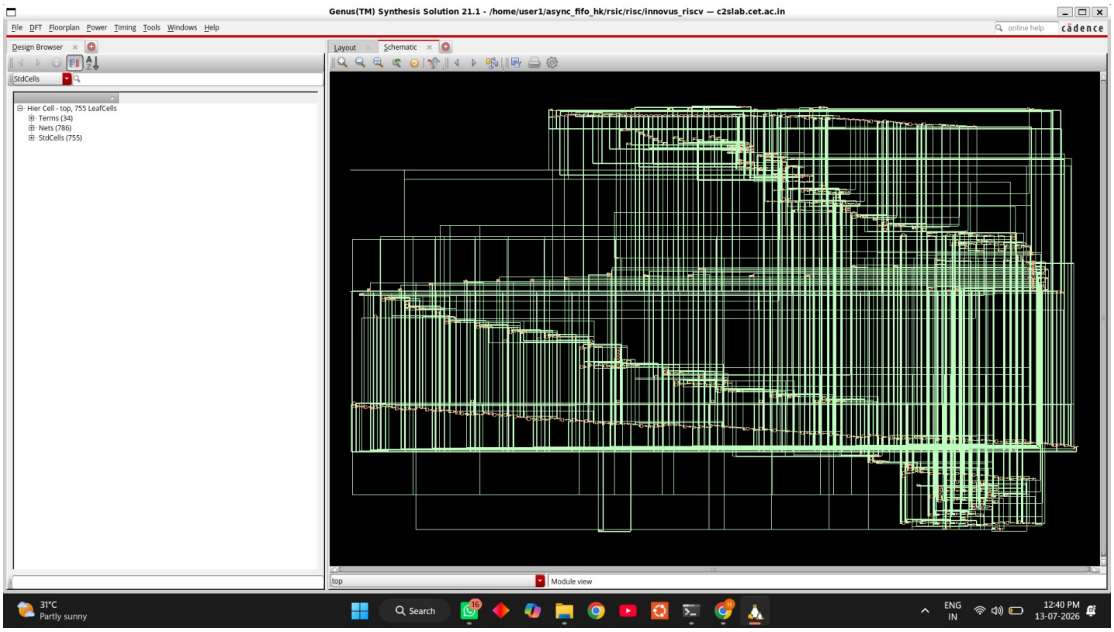


Figure 4.2: Logic Synthesis Using Cadence Genus

The synthesis stage completed successfully without any unresolved references or synthesis errors. Various synthesis reports containing information related to timing performance, area utilization, design hierarchy, and standard-cell usage were generated for analysis. The successful generation of these reports confirmed that the RTL description had been accurately converted into a technology-mapped gate-level implementation while satisfying the specified design constraints.

The generated gate-level netlist represents the synthesized implementation of the processor using the target standard-cell library. The successful completion of logic synthesis confirmed that the design was suitable for post-synthesis verification and the subsequent physical design stages.

4.4 Verification of the Synthesized Design

Before proceeding to the physical design stage, the synthesized gate-level netlist was verified to ensure that the logic synthesis process had preserved the functionality of the original RTL design. Two verification techniques, namely Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS), were employed to validate the synthesized implementation.

Cadence Conformal was used to perform Logic Equivalence Checking by comparing the synthesized gate-level netlist with the original RTL description.

The LEC process completed successfully with a PASS status, confirming that the synthesized netlist remained functionally equivalent to the original RTL design. This

result verified that the logic synthesis process preserved the intended functionality of the processor.

Gate-Level Simulation (GLS) was subsequently carried out using the synthesized netlist together with the same verification environment used during RTL simulation. The generated simulation waveforms matched the RTL simulation results, confirming that the synthesized implementation behaved exactly as intended. The successful completion of both LEC and GLS established that the synthesized design was ready for physical implementation.

4.5 Physical Design Results

The verified synthesized netlist was imported into Cadence Innovus to begin the physical design stage. During this stage, the logical design was transformed into a manufacturable physical layout through a sequence of implementation steps, including floorplanning, power planning, placement, Clock Tree Synthesis (CTS), and routing.

4.5.1 Floorplanning Results

The first stage of physical implementation involved floorplanning, where the overall chip dimensions, core region, placement rows, routing resources, and I/O pad locations were defined. Appropriate values for the aspect ratio and core utilization were selected to achieve efficient silicon utilization while providing sufficient routing resources for the design.

Figure 4.3 illustrates the generated floorplan.

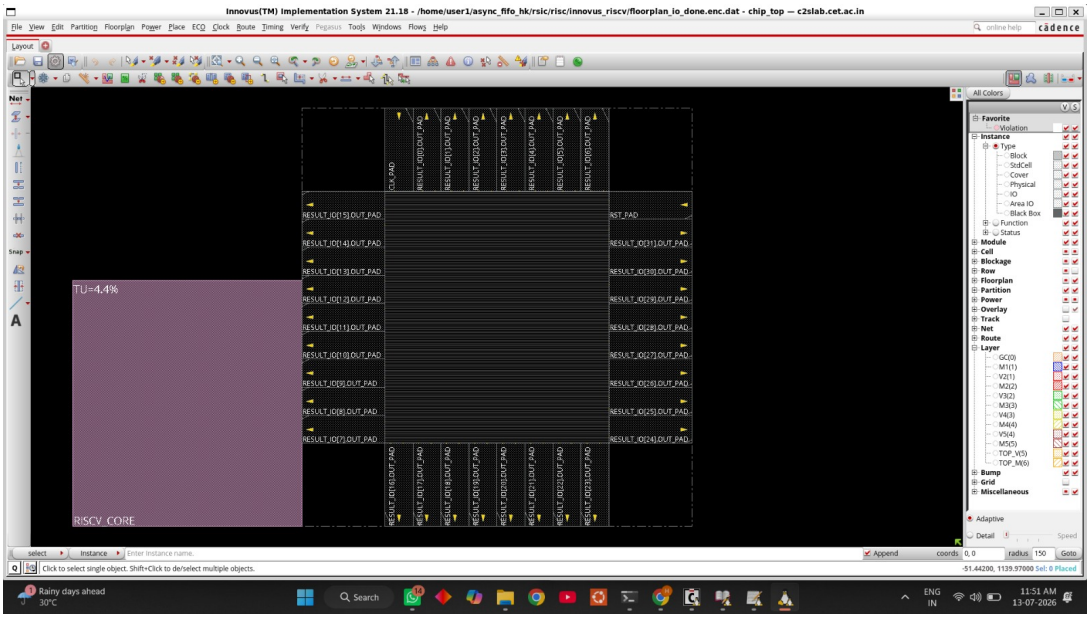


Figure 4.3: Floorplan

The generated floorplan established the physical boundaries of the processor and served as the foundation for the remaining implementation stages. An optimized floorplan reduced routing complexity, improved area utilization, and provided adequate space for standard-cell placement and interconnect routing.

4.5.2 Power Planning Results

Following floorplanning, a robust power distribution network was created to ensure reliable power delivery throughout the processor. Power rings surrounding the core together with horizontal and vertical power stripes were implemented to distribute the supply voltage uniformly across the design.

Figure 4.4 shows the power planning result.

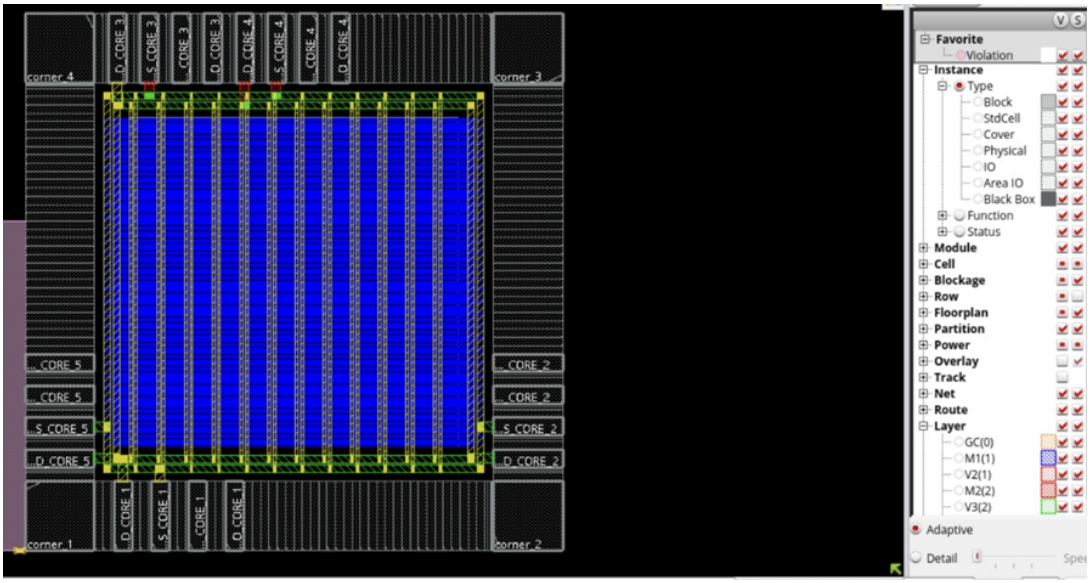


Figure 4.4: Power Planning

The completed power network provided stable voltage distribution across the chip while minimizing voltage drop and improving overall reliability. A well-designed power distribution network is essential for maintaining stable circuit operation during physical implementation.

4.5.3 Placement Results

Following the completion of power planning, the synthesized standard cells were placed within the core region using Cadence Innovus. During placement, the tool optimized the location of each standard cell to reduce interconnect length, minimize routing congestion, and improve timing performance while satisfying all placement constraints.

Figure 4.5 illustrates the placement result.

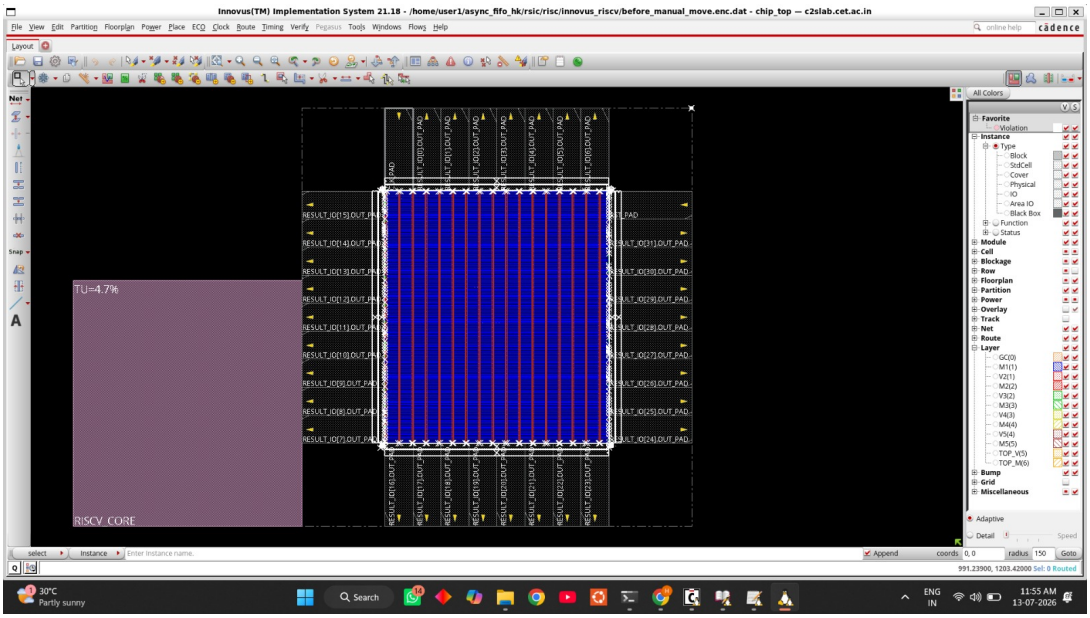


Figure 4.5: Placement of Standard Cells

The placement stage completed successfully, with the standard cells distributed efficiently throughout the core area. The optimized placement minimized congestion, improved routing efficiency, and established a suitable foundation for Clock Tree Synthesis and routing. In addition, efficient placement contributed to improved timing performance and reduced interconnect delay.

4.5.4 Clock Tree Synthesis Results

After the placement stage, Clock Tree Synthesis (CTS) was performed to establish an efficient clock distribution network for the processor. During this process, Cadence Innovus automatically inserted clock buffers and inverters to balance the clock paths, thereby minimizing clock skew and insertion delay across all sequential elements.

Figure 4.6 illustrates the generated clock tree.

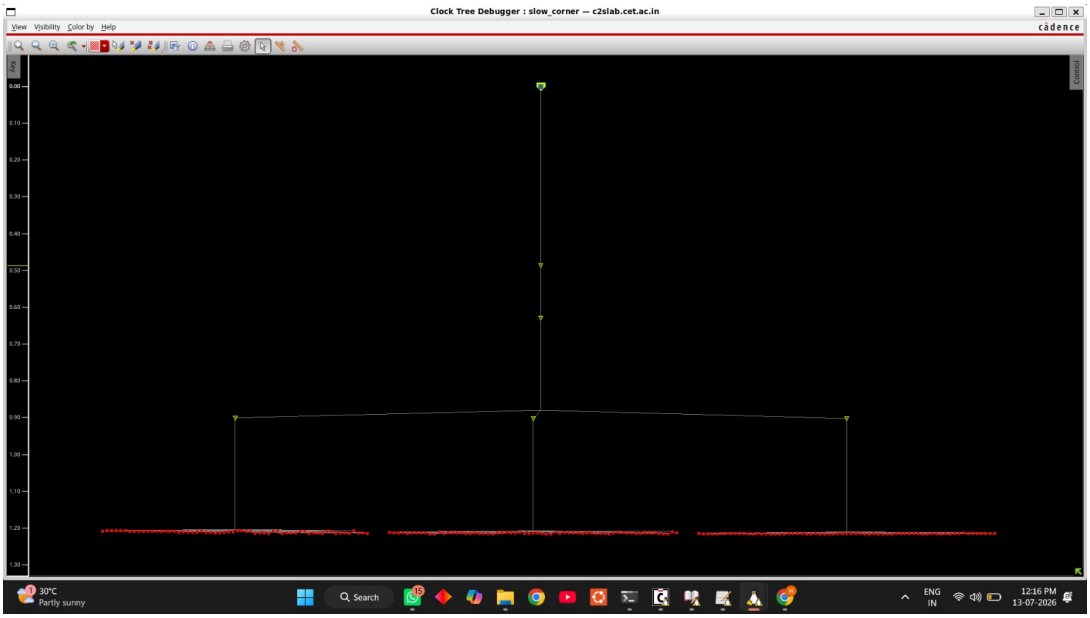


Figure 4.6: Clock Tree Synthesis

The generated clock network provided balanced clock distribution throughout the design, ensuring reliable synchronization of all sequential components. The successful completion of CTS minimized clock-related timing issues and prepared the design for the routing stage.

4.5.5 Routing Results

Following Clock Tree Synthesis, the routing stage was carried out to establish all signal, clock, and power interconnections within the processor. Cadence Innovus first executed global routing to determine suitable routing paths and subsequently performed detailed routing to generate the final metal interconnections while satisfying the design rules of the target technology.

The routing stage was completed successfully without significant congestion or connectivity issues. All signal, clock, and power connections were established correctly while complying with the routing constraints of the selected technology. The successful completion of routing marked the realization of the complete physical layout and prepared the design for timing analysis and physical verification. Figure 4.7 illustrates the generated clock tree.

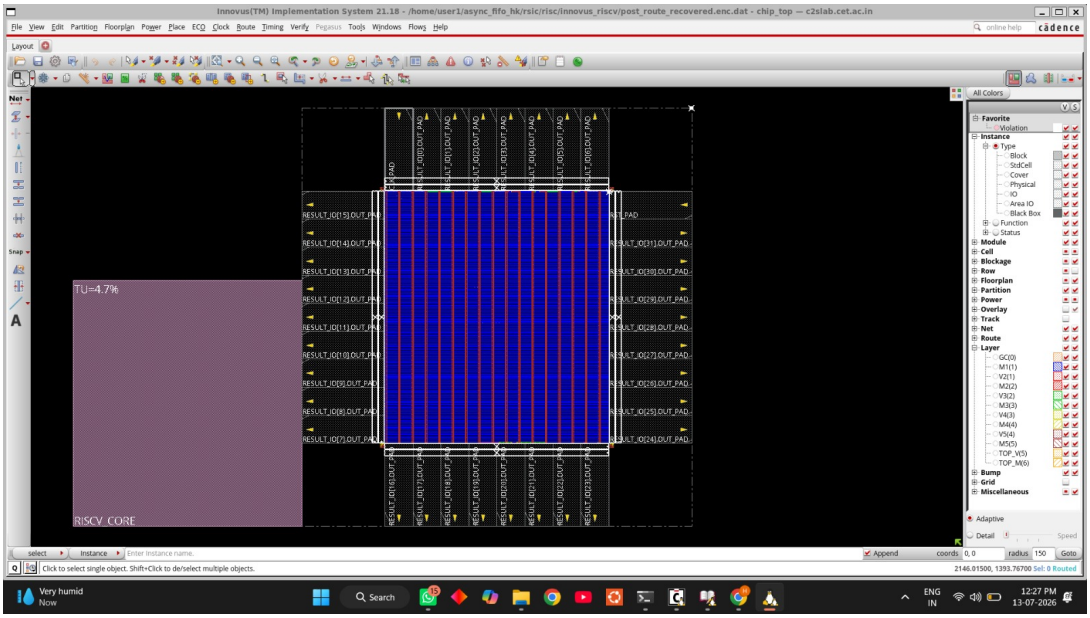


Figure 4.7: Clock Tree Synthesis

4.6 Timing Analysis and Timing Signoff Results

Following the routing stage, Static Timing Analysis (STA) was carried out using Cadence Tempus to evaluate the timing characteristics of the implemented processor. Timing reports were generated to verify that the design satisfied the required setup and hold timing constraints under the specified operating conditions.

Comprehensive timing reports were generated to analyze the critical timing paths of the processor.

```
   *** timeDesign #1 [begin] : totSession cpu/real = 0:01:43.2/0:10:43.1 (0.2), mem =
2221.2M Start to check current routing status for nets... All nets are already routed
correctly. End to check current routing status for nets (mem=2179.2M)

   ------------------------------------------------------------------       timeDesign
Summary ------------------------------------------------------------------

   Setup views included: slow_view

   +--------------------+---------+---------+---------+ | Setup mode | all | reg2reg
| default | +--------------------+---------+---------+---------+ | WNS (ns):| 0.007 |
0.108 | 0.007 | | TNS (ns):| 0.000 | 0.000 | 0.000 | | Violating Paths:| 0 | 0 | 0 |
| All Paths:| 320 | 288 | 32 | +--------------------+---------+---------+---------+

   +----------------+--------------------------------+------------------+ |  | Real  |
Total   |   |   DRVs   +------------------+------------+------------------+  |   |   |   Nr
nets(terms) | Worst Vio | Nr nets(terms) | +----------------+------------------+-----
-------+------------------+ | max_cap | 0 (0) | 0.000 | 0 (0) | | max_tran | 0 (0) |
0.000 | 0 (0) | | max_fanout | 0 (0) | 0 | 0 (0) | | max_length | 0 (0) | 0 | 0 (0) |
+----------------+------------------+------------+------------------+

   Density: 4.717% Routing Overflow: 0.01% H and 0.00% V ----------------------------
---------------------------------------- Reported timing to dir ./timingReports Total
CPU time: 0.45 sec Total Real time: 1.0 sec Total Memory Usage: 2179.339844 Mbytes
```

Figure 4.8: Detailed Timing Report

The timing reports confirmed that the implemented design satisfied all setup and hold timing requirements. Timing closure was achieved successfully, demonstrating that the processor could operate reliably at the intended clock frequency without violating timing constraints. This result confirmed that the physical implementation satisfied the required timing specifications.

4.7 Physical Design Verification Results

Following timing signoff, the completed layout underwent physical verification to ensure that it satisfied both functional and fabrication requirements. Post-Route Logic Equivalence Checking (LEC) was first performed to verify that the optimizations introduced during physical implementation had not altered the logical functionality of the processor.

The verification completed successfully with a PASS status, confirming that the routed implementation remained logically equivalent to the synthesized netlist. This verified that the logical behaviour of the processor was preserved throughout the physical design process.

The final physical layout generated after routing is shown in Figure 4.9.

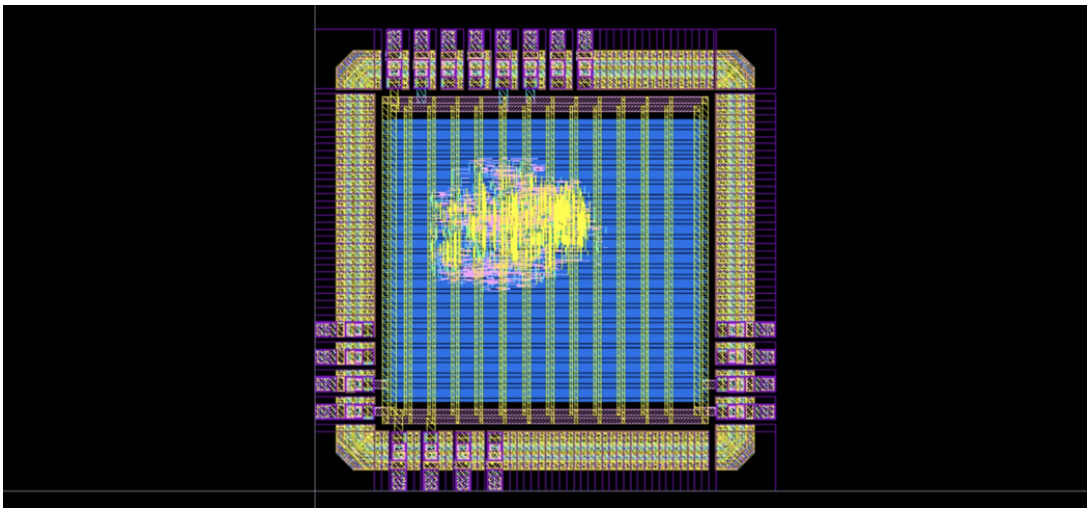


Figure 4.9: Final Physical Layout

The completed layout represents the successful implementation of the Single-Cycle RISC-V Processor through the complete RTL-to-GDSII design flow. It includes the placed standard cells, clock distribution network, routed interconnections, power delivery network, and I/O structures required for fabrication.

Design Rule Checking (DRC) was subsequently performed to verify compliance with the fabrication rules of the target technology.

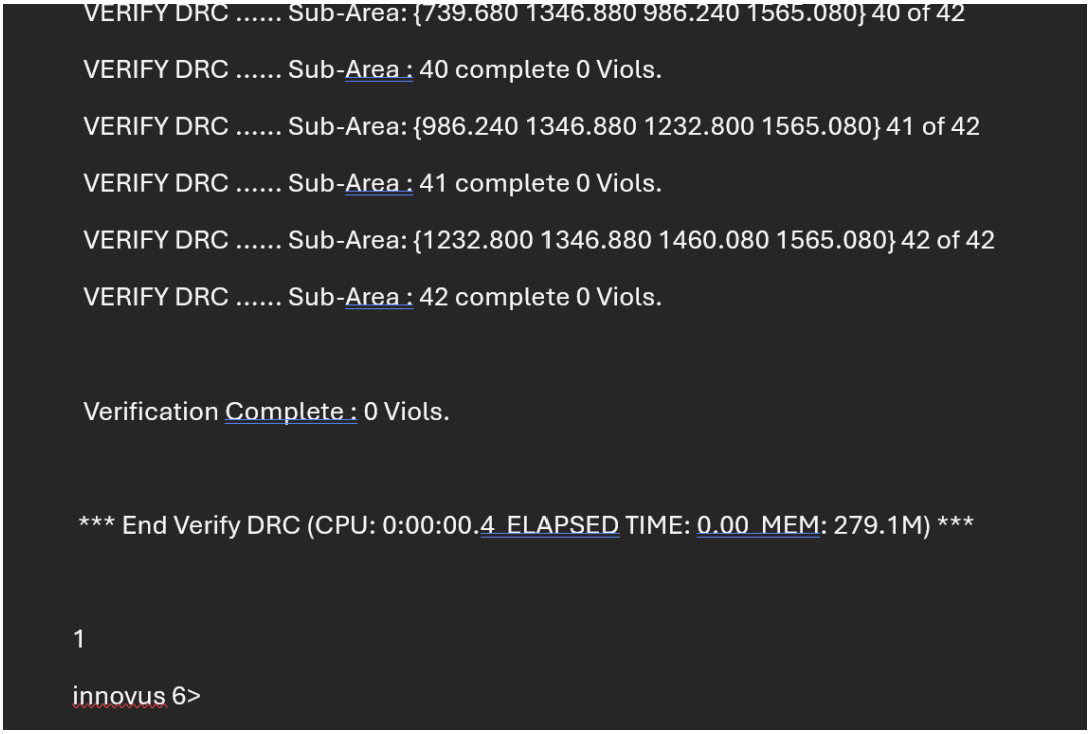


Figure 4.10: Design Rule Check (DRC) Result

The DRC process completed successfully without any rule violations, confirming that the layout complied with all manufacturing requirements. The successful completion of Post-Route LEC and DRC verified that the implemented processor was functionally correct, physically valid, and ready for semiconductor fabrication.

4.8 Overall Implementation Summary

The implementation of the Single-Cycle RISC-V Processor was successfully completed by following the complete RTL-to-GDSII ASIC design flow. Each implementation stage, beginning with RTL modelling and functional validation, followed by logic synthesis, post-synthesis verification, physical design, timing analysis, and physical verification, was completed successfully. The results obtained throughout the implementation confirmed the functional correctness of the processor and demonstrated the successful realization of the design using industry-standard Cadence EDA tools.

Table 4.1: Summary of Implementation Results

Implementation Stage	**Status**
RTL Simulation and Functional Validation	Successful
Technology Mapping through Logic Synthesis	Successful
Post-Synthesis Logic Equivalence Checking	PASS
Gate-Level Simulation	Successful
Floorplanning	Completed Successfully
Power Planning	Completed Successfully
Placement	Completed Successfully
Clock Tree Synthesis	Completed Successfully
Routing	Completed Successfully
Static Timing Analysis and Timing Signoff	Timing Closure Achieved
Post-Route Logic Equivalence Checking	PASS
Design Rule Checking (DRC)	Passed (No Violations)
Final GDSII Layout Generation	Successful

Chapter 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This internship successfully accomplished the complete RTL-to-GDSII implementation of a Single-Cycle RISC-V Processor using industry-standard Cadence Electronic Design Automation (EDA) tools. The implementation began with the development of the Register Transfer Level (RTL) model in Verilog Hardware Description Language (HDL) and progressed through functional validation, logic synthesis, physical design, timing analysis, and physical verification before generating the final GDSII layout for semiconductor fabrication.

The functional behaviour of the processor was first validated through RTL simulation, confirming the correct execution of the supported RISC-V instruction set. Logic synthesis successfully converted the RTL description into a technology-mapped gate-level netlist. The correctness of the synthesized implementation was further established using Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS), which confirmed that the synthesized design preserved the functionality of the original RTL implementation. The physical design was then completed through floorplanning, power planning, placement, Clock Tree Synthesis (CTS), and routing using Cadence Innovus.

Static Timing Analysis (STA) performed using Cadence Tempus verified that the implemented processor satisfied all specified setup and hold timing requirements, resulting in successful timing closure. Physical verification, including Post-Route Logic Equivalence Checking and Design Rule Checking (DRC), confirmed that the final layout maintained functional correctness while complying with the manufacturing rules of the selected semiconductor technology. The successful completion of every stage demonstrated the effec-

tiveness of the standard RTL-to-GDSII ASIC implementation flow and confirmed that the processor was ready for fabrication.

In addition to the successful implementation of the processor, the internship provided valuable hands-on experience with the complete ASIC design flow followed in the semiconductor industry. It strengthened the understanding of digital VLSI design, processor architecture, logic synthesis, physical design, timing analysis, verification techniques, and the practical use of professional Cadence EDA tools. The knowledge and experience gained through this internship establish a strong foundation for future work in ASIC design, VLSI engineering, and semiconductor research.

5.2 Future Scope

The successful implementation of the Single-Cycle RISC-V Processor provides a strong foundation for further research and development in ASIC design. Although the objectives of this internship were achieved, several enhancements can be incorporated in future work to improve the processor's performance, functionality, and overall efficiency.

- Design and implement a pipelined RISC-V processor to increase instruction throughput and overall processing performance.
- Extend the processor by incorporating additional RISC-V ISA extensions, such as multiplication and division (M), compressed instructions (C), and floating-point operations (F).
- Integrate instruction and data cache memories to reduce memory access latency and enhance execution efficiency.
- Explore advanced low-power design techniques to minimize both dynamic and leakage power consumption.
- Evaluate the implementation using advanced semiconductor technology nodes and compare improvements in performance, power, and silicon area.
- Further optimize the physical design to achieve higher operating frequencies while maintaining timing closure.
- Incorporate standard peripheral interfaces to support embedded system and System-on-Chip (SoC) applications.
- Fabricate the processor using an appropriate semiconductor technology and validate its functionality through post-fabrication testing.

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Appendix A

WORK PLAN

The internship was conducted over a duration of eight weeks under the Chips-to-Startup (C2S) Programme. The activities completed during each week are summarized in Table A.1.

Table A.1: Weekly Work Plan

Week	**Activities Completed**
1	Familiarization with the RTL-to-GDSII ASIC implementation flow, Linux environment, Cadence EDA tools, semiconductor fabrication concepts, and the architecture of the RISC-V processor.
2	Development and individual verification of Verilog modules such as the Program Counter, Register File, Instruction Memory, Data Memory, Arithmetic Logic Unit (ALU), Control Unit, Immediate Generation Unit, multiplexers, and other supporting modules.
3	Integration of the developed RTL modules to build the complete Single-Cycle RISC-V Processor followed by functional validation using a Verilog testbench.
4	Logic synthesis of the RTL design using Cadence Genus followed by post-synthesis validation through Logic Equivalence Checking (LEC) and Gate-Level Simulation (GLS).
5	Execution of the initial physical design stages, including floorplanning, power planning, placement, and physical optimization using Cadence Innovus.
6	Implementation of Clock Tree Synthesis (CTS), routing, timing optimization, and generation of the final routed layout.
7	Static Timing Analysis (STA), Post-Route Logic Equivalence Checking (LEC), Design Rule Checking (DRC), and physical verification of the implemented design.
8	Generation of the final GDSII layout, evaluation of implementation results, preparation of technical documentation, and compilation of the final internship report.